

CHERNIVTSI, Ukraine

WMO: 336580

Lat: 48.2664N

Lon: 25.9725E

Elev: 246

StdP: 98.40

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.5	-19.1	0.7	-16.1	-15.9	1.0	-12.9	11.5	-0.4	10.2	-2.1	2.6	110	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	30.8	20.4	29.0	19.8	27.3	19.0	22.0	27.8	21.0	26.8	20.1	25.5	3.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.2	25.2	19.0	14.3	23.8	18.1	13.5	22.7	65.6	28.0	61.8	26.8	58.6	25.6	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.1	7.2	DB	-20.3	33.4	3.7	2.0	-22.9	34.8	-25.1	36.0	-27.1	37.1	-29.8	38.5	(4)
(5)			WB	-20.5	23.8	3.6	1.2	-23.1	24.6	-25.3	25.3	-27.3	26.0	-30.0	26.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-3.0	-1.0	3.7	10.1	15.4	18.9	20.8	20.1	15.1	9.3	3.9	-1.5	(6)	
	DBStd	9.45	5.54	5.75	4.94	4.17	3.59	3.16	2.83	3.03	3.55	4.23	4.54	5.27	(7)	
	HDD10.0	1578	403	309	202	47	3	0	0	0	3	64	190	357	(8)	
	HDD18.3	3502	661	542	455	247	103	32	10	16	108	279	434	615	(9)	
	CDD10.0	1346	0	0	4	51	172	267	333	313	156	43	6	0	(10)	
	CDD18.3	228	0	0	0	1	12	48	85	71	11	1	0	0	(11)	
	CDH23.3	1878	0	0	0	17	133	375	661	575	111	6	0	0	(12)	
	CDH26.7	508	0	0	0	3	19	84	193	187	22	0	0	0	(13)	
(14) Wind	WSAvg	3.1	3.1	3.4	3.7	3.7	3.2	3.1	2.9	2.6	2.8	2.9	3.1	3.1	(14)	
(15) Precipitation	PrecAvg	653	32	30	34	53	73	94	101	60	54	39	37	37	(15)	
	PrecMax	888	182	77	134	136	185	260	232	129	175	207	113	91	(16)	
	PrecMin	390	1	2	6	11	15	0	24	6	1	2	10	6	(17)	
	PrecStd	132	30	18	23	29	40	50	56	32	47	35	22	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	14.2	19.2	25.3	28.8	30.9	33.0	33.2	29.2	24.0	18.2	12.0	(19)	
		MCWB	7.1	8.1	10.7	14.8	18.1	20.4	21.1	20.4	18.5	16.4	12.1	7.4	(20)	
	2%	DB	7.7	10.8	15.9	22.0	26.7	29.0	30.8	30.9	26.2	20.7	14.8	9.0	(21)	
		MCWB	4.9	6.8	9.4	13.3	17.0	19.8	20.8	20.3	17.0	14.7	10.8	6.0	(22)	
	5%	DB	5.4	8.2	13.1	19.8	24.8	27.2	28.9	28.8	24.0	17.9	12.2	6.2	(23)	
		MCWB	3.1	5.1	7.9	12.2	16.5	19.1	20.4	19.8	16.7	13.1	9.5	4.4	(24)	
	10%	DB	3.9	5.9	10.8	17.4	22.6	25.6	27.0	26.7	21.7	15.7	10.2	4.6	(25)	
		MCWB	2.4	3.7	6.7	11.2	15.6	18.5	19.4	19.1	15.7	12.1	8.3	3.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	8.8	11.5	15.7	20.2	22.7	23.4	23.1	20.2	17.2	12.7	8.1	(27)	
		MCDB	9.9	13.3	17.3	23.1	26.6	27.6	30.3	29.5	26.8	22.7	17.0	11.3	(28)	
	2%	WB	5.1	7.1	9.8	14.3	18.2	21.1	22.1	21.8	18.3	15.1	11.2	6.2	(29)	
		MCDB	6.9	10.2	14.5	20.9	24.1	26.7	28.1	27.7	23.8	19.3	13.8	8.3	(30)	
	5%	WB	3.7	5.4	8.4	12.9	17.1	20.1	21.1	20.7	17.2	13.8	9.9	4.7	(31)	
		MCDB	5.0	7.8	12.4	18.2	22.7	25.4	26.7	26.5	22.1	17.2	11.9	6.2	(32)	
	10%	WB	2.3	3.9	7.1	11.7	16.1	19.2	20.1	19.6	16.2	12.5	8.5	3.3	(33)	
		MCDB	3.6	5.7	10.1	16.4	21.2	24.2	25.4	24.9	20.6	15.2	10.2	4.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.3	7.9	9.9	10.3	9.6	9.8	10.2	9.4	8.5	6.0	5.2	(35)	
		MCDBR	7.4	9.8	12.7	14.0	13.9	12.7	13.1	14.0	14.0	12.4	9.8	8.1	(36)	
	5% WB	MCWBR	5.6	6.6	7.4	6.9	6.2	5.3	5.2	5.2	6.1	7.0	6.3	5.8	(37)	
		MCWBR	6.6	9.3	11.7	12.3	12.0	11.1	11.3	12.1	11.8	11.0	8.4	7.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.313	0.343	0.382	0.440	0.415	0.413	0.442	0.448	0.404	0.368	0.339	0.310	(40)	
		taud	2.281	2.234	2.219	2.105	2.248	2.282	2.214	2.192	2.271	2.348	2.369	2.344	(41)	
	Ebn at Noon		738	796	823	808	848	850	818	793	793	762	706	696	(42)	
		Edh at Noon	75	99	118	146	132	128	136	133	112	88	70	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.08	1.87	3.01	4.28	5.35	5.67	5.58	5.03	3.54	2.27	1.12	0.81	(44)	
	RadStd		0.15	0.26	0.36	0.42	0.49	0.45	0.44	0.42	0.50	0.31	0.13	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.67	N/A	N/A	+1.02	N/A	N/A	N/A	-228	+152	+62	(46)
(47)	Regional (25 neighbors)	+0.72	N/A	N/A	+0.94	N/A	N/A	-153	-217	+107	+34	(47)

Nomenclature: See separate page